

Let, $a=c$ and $b=d$. By Defn. 2, $\{a\}=\{c\}$ and $\{a,b\}=\{c,d\}$.

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$\therefore (a,b) = (c,d) \quad \square$

50) This exercise presents Russell's paradox. Let S be the set that contains a set x if the set x does not belong to itself, so that $S = \{x \mid x \notin x\}$.

- Show the assumption that S is a member of S leads to a contradiction.
- Show the assumption that S is not a member of S leads to a contradiction.

Solution: a) Let $S \in S$. By definition, $\because S$ is a member of S , $\therefore S \notin S$.

But this is a contradiction

b) Let $S \notin S$. By definition, $\therefore S \in S$. But this is a contradiction. $\square$

By parts (a) and (b) it follows that the set S cannot be defined as it was. This paradox can be avoided by restricting the types of elements that sets can have.

51) Describe a procedure for listing all the subsets of a finite set.

Solution: Let, S be a finite set containing n distinct elements. $\therefore |S| = n$.

Rename the elements of S as $a_1, a_2, \dots, a_n$. By Defn. 6, $\mathcal{P}(S)$ is the set of all subsets of S . We claim that each of the n length bit strings corresponds to a subset of S and hence is a member of $\mathcal{P}(S)$.
 $\forall i, 1 \leq i \leq n$, $a_i \in$ the given subset implies its corresponding position is 1 in the bit string. If $a_i \notin$ the given subset implies its corresponding position is 0 in the bit string.

Example: The set of vowels $V = \{a, e, i, o, u\}$ Renaming: $a \rightarrow a_1, e \rightarrow a_2, i \rightarrow a_3, o \rightarrow a_4, u \rightarrow a_5$

00000 $\rightarrow \emptyset$	01000 $\rightarrow \{e\}$	10000 $\rightarrow \{a\}$	11011 $\rightarrow \{a, e, o, u\}$
00001 $\rightarrow \{u\}$	01001 $\rightarrow \{e, u\}$	10001 $\rightarrow \{a, u\}$	11100 $\rightarrow \{a, e, i\}$
00010 $\rightarrow \{o\}$	01010 $\rightarrow \{e, o\}$	10010 $\rightarrow \{a, o\}$	11101 $\rightarrow \{a, e, i, u\}$
00011 $\rightarrow \{o, u\}$	01011 $\rightarrow \{e, o, u\}$	10011 $\rightarrow \{a, o, u\}$	11110 $\rightarrow \{a, e, i, o\}$
00100 $\rightarrow \{i\}$	01100 $\rightarrow \{e, i\}$	10100 $\rightarrow \{a, i\}$	11111 $\rightarrow \{a, e, i, o, u\}$
00101 $\rightarrow \{i, u\}$	01101 $\rightarrow \{e, i, u\}$	10101 $\rightarrow \{a, i, u\}$	
00110 $\rightarrow \{i, o\}$	01110 $\rightarrow \{e, i, o\}$	10110 $\rightarrow \{a, i, o\}$	
00111 $\rightarrow \{i, o, u\}$	01111 $\rightarrow \{e, i, o, u\}$	10111 $\rightarrow \{a, i, o, u\}$	
		11000 $\rightarrow \{a, e\}$	
		11001 $\rightarrow \{a, e, u\}$	
		11010 $\rightarrow \{a, e, o\}$	

$\square$